

Permutations and Combinations

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COMP2711 Discrete Mathematical Tools

1 Ordered and unordered outcomes

Select three of six students to form an ICPC team. Compare this with selecting three students and arranging them in a photograph.

- In a photograph, ABC and BAC are different arrangements.
- In a team, $\{A, B, C\}$ is the same set no matter how the names are written.

MUST UNDERSTAND Ask first: does swapping two selected objects change the outcome?

	Permutation	Combination
outcome	ordered list	subset
swapping two choices	changes the outcome	does not change it

We first count ordered selections, then remove order using the division rule, and finally allow repeated types. Restrictions are handled with the product, sum, and complement rules from the previous lecture.

2 Permutations

2.1 Factorials

Arrange A, B, C, D in a photograph. Fill the positions from left to right:

Position	People available	Choices	Example choice
1	A, B, C, D	4	B
2	A, C, D	3	C
3	A, D	2	A
4	D	1	D

After each choice, one fewer person remains. By the product rule, there are $4 \cdot 3 \cdot 2 \cdot 1 = 24$ photographs.

For an integer $n \geq 1$,

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1, \quad 0! = 1.$$

The value $n!$ counts the ways to arrange n distinct objects in a line. The definition $0! = 1$ counts the single empty arrangement.

2.2 Selecting and arranging

MUST UNDERSTAND A permutation is an ordered arrangement of distinct objects.

For example, choose four of six people A, B, C, D, E, F and arrange them in a photograph. The four positions have 6, 5, 4, 3 choices, giving $P(6, 4) = 6 \cdot 5 \cdot 4 \cdot 3 = 360$ photographs.

In general, select and arrange r of n distinct objects. The successive positions have

$$n, \quad n-1, \quad n-2, \quad \dots, \quad n-r+1$$

choices. Therefore

$$P(n, r) = n(n-1) \cdots (n-r+1) = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n.$$

Read $P(n, r)$ as “ordered selections of r distinct objects from n .” For $r = 0$, the product has no factors and equals 1.

Why divide by $(n-r)!$? Arrange all n people, then crop the photograph to its first r positions. For each fixed cropped photograph, the remaining $n-r$ people have $(n-r)!$ orders, all producing the same cropped result. For example,

$$\begin{array}{cccc|cc} B & C & A & D & E & F \\ B & C & A & D & F & E \end{array}$$

same photograph cropped out

Every cropped photograph is counted exactly $(n-r)!$ times among the $n!$ full arrangements. The division rule gives $P(n, r) = n!/(n-r)!$.

2.3 Awards

Eight finalists compete for gold, silver, and bronze, with no ties. The awards are distinct positions, so

$$P(8, 3) = 8 \cdot 7 \cdot 6 = 336.$$

The expression $\binom{8}{3}$ would choose only the three medallists and would forget which medal each person receives.

2.4 Fixed blocks

How many permutations of **ABCDEFGH** contain the consecutive block **ABC** in that order?

Treat **ABC** as one object. The six objects

$$\mathbf{ABC}, D, E, F, G, H$$

can be arranged in

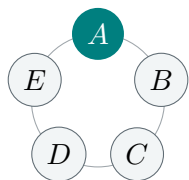
$$6! = 720$$

ways. If the three letters only had to stay together in any internal order, we would multiply by $3!$.

2.5 Circular arrangements

Seat $n \geq 1$ distinct people around a round table with unlabelled seats. Rotations count as the same seating. Rotate every arrangement so that A is at the top; the other $n - 1$ people can occupy the clockwise positions in $(n - 1)!$ ways. Thus the number of circular arrangements is

$$\frac{n!}{n} = (n - 1)!.$$



Fix A ; arrange the other four people.

Reflected orders count as different. For $n \geq 3$, a reflection reverses the clockwise order and gives a different seating.

3 Combinations

Choose three people from A, B, C, D, E , with no roles or ordering. Counting ordered selections gives $P(5, 3) = 60$, but the team $\{A, B, C\}$ appears as

$$ABC, \quad ACB, \quad BAC, \quad BCA, \quad CAB, \quad CBA.$$

Every three-person team has exactly $3! = 6$ descriptions. Dividing gives $60/6 = 10$ teams.

MUST UNDERSTAND A combination is an unordered selection: an r -element subset.

First count ordered selections. Every r -element subset has exactly $r!$ orders, so

$$P(n, r) = \binom{n}{r} r!.$$

Dividing by the uniform $r!$ descriptions gives

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n - r)!}, \quad 0 \leq r \leq n.$$

Read $\binom{n}{r}$ as “choose r objects from n , without order and without repetition.”

3.1 A five-country council

Use a pool of 193 UN member states. Select five distinct countries for a *hypothetical* advisory council, with no eligibility restrictions and the same role for every member. This is a subset selection:

$$\begin{aligned}\binom{193}{5} &= \frac{193!}{5!188!} = \frac{193 \cdot 192 \cdot 191 \cdot 190 \cdot 189}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \\ &= 2,118,000,528.\end{aligned}$$

The numerator counts ordered selections; the denominator removes the $5!$ orders of each council. This is not the selection procedure for an actual UN body.¹

3.2 School IDs

An ID has five characters, each an uppercase letter A–Z or a digit 0–9, and contains at most two letters. Characters may repeat and leading zeroes are allowed. Examples include 01234, 7A203, B4C09, and AA007.

Split into the disjoint cases of zero, one, or two letters. With exactly k letters, first choose their k positions, then fill them with letters and fill the remaining positions with digits:

$$\binom{5}{k} 26^k 10^{5-k}.$$

For example, letter positions $\{2, 4\}$ give the pattern digit–letter–digit–letter–digit. The positions form a subset, so there are $\binom{5}{2}$ choices, not $P(5, 2)$.

Letters	Position choices	Count	Value
0	$\binom{5}{0} = 1$	10^5	100,000
1	$\binom{5}{1} = 5$	$5 \cdot 26 \cdot 10^4$	1,300,000
2	$\binom{5}{2} = 10$	$10 \cdot 26^2 \cdot 10^3$	6,760,000

By the sum rule, the total is $100,000 + 1,300,000 + 6,760,000 = 8,160,000$.

3.3 Complement identity

For integers $0 \leq r \leq n$, choosing r elements to include determines the $n - r$ elements to leave out, and conversely. Therefore

$$\binom{n}{r} = \binom{n}{n-r}.$$

The factorial formulas also show the equality:

$$\binom{n}{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{(n-r)!r!} = \binom{n}{r}.$$

¹Membership reference: [United Nations](#), [Main Bodies](#).

Example. Choose 18 of 20 students for a trip. Instead, choose the two who stay out. Both choices give the same count:

$$\binom{20}{18} = \binom{20}{2} = \frac{20!}{18!2!} = \frac{20 \cdot 19 \cdot \cancel{18!}}{\cancel{18!} \cdot 2 \cdot 1} = 190.$$

Cancel the common factorial before multiplying.

3.4 Subsets and bit strings

Let $S = \{s_1, \dots, s_n\}$. Encode a subset with a length- n bit string: bit i is 1 exactly when s_i is selected. A subset has size r exactly when its bit string has r ones. Therefore

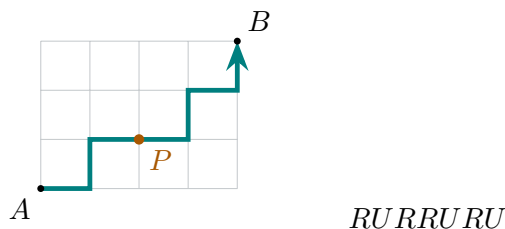
$$\#\{\text{length-}n \text{ bit strings with } r \text{ ones}\} = \binom{n}{r}.$$

For example, $\{s_2, s_4, s_5\} \subseteq \{s_1, \dots, s_6\}$ corresponds to 010110.

3.5 Paths on a grid

How many paths go from $A = (0, 0)$ to $B = (4, 3)$ using only unit steps right or up? Every path has four right steps and three up steps. Choose the three up-step positions among seven positions; all remaining steps are right steps. This correspondence works in both directions, so the count is

$$\binom{7}{3} = 35.$$



If the path must pass through $P = (2, 1)$, count its two segments separately. The first has two right steps and one up step; the second has two of each:

$$\binom{3}{1} \binom{4}{2} = 3 \cdot 6 = 18.$$

Consequently, $35 - 18 = 17$ paths avoid P . The two-segment product counts each path through P once, since the split at P is unique.

4 Combinations with repetition

4.1 Six cards, four suits

Draw six cards from a standard 52-card deck without replacement. After each draw, put the card in its suit's location. Ignore ranks and order: an outcome records only the number of cards in each suit.

Keep the suit order $\heartsuit, \spadesuit, \diamondsuit, \clubsuit$. For example, $(3, 0, 2, 1)$ means three hearts, no spades, two diamonds, and one club.



Why not divide 4^6 by $6!$? There are 4^6 possible suit sequences, but their overcount is not uniform. Six hearts have just one sequence. Five hearts and one spade have six sequences, because the spade can occupy any of the six positions. We need a representation that describes each suit-count list exactly once.

4.2 Stars and bars

Use one star for each card and three bars to separate the four suits. The stars before bar 1 represent $\heartsuit$; between bars 1 and 2, $\spadesuit$; between bars 2 and 3, $\diamondsuit$; and after bar 3, $\clubsuit$.

Counts ($\heartsuit, \spadesuit, \diamondsuit, \clubsuit$)	Pattern
$(3, 0, 2, 1)$	$*** ** *$
$(1, 2, 1, 2)$	$* ** * **$
$(6, 0, 0, 0)$	$***** $

Adjacent bars allow a suit count of zero. A bar at an end allows the first or last count to be zero.

Six stars and three bars occupy nine slots. Once we choose the three bar positions, all remaining slots contain stars:



Every suit-count list gives exactly one bar placement, and every placement gives exactly one list. Hence

$$\text{number of suit-count lists} = \binom{9}{3} = \frac{9 \cdot 8 \cdot 7}{3 \cdot 2 \cdot 1} = 84.$$

These are suit counts, not individual six-card hands. All these counts are possible because no suit needs more than six of its 13 cards.

4.3 The general formula

MUST UNDERSTAND A combination with repetition selects quantities from types; order is ignored.

Choose r objects from n types, each with unlimited supply, where $n \geq 1$ and $r \geq 0$ are integers. Equivalently, assign a non-negative integer count x_i to each type i , satisfying

$$x_1 + \cdots + x_n = r, \quad x_i \geq 0.$$

Use r stars and $n - 1$ bars in $r + n - 1$ slots. Choosing the bar positions gives

$$\binom{r + n - 1}{n - 1} = \binom{r + n - 1}{r}.$$

The second expression chooses the star positions instead. **Remember the -1 :** $n - 1$ bars are enough to separate n types. For $n = 1$, there are no bars and just one possible quantity, $x_1 = r$. If order matters instead, there are n^r sequences; that is a different problem.

4.4 Nondecreasing digits

How many six-digit decimal numbers have nondecreasing digits? Zero cannot appear because it would be the first digit. Choose six digits with repetition from $\{1, \dots, 9\}$; their order is then forced:

$$\binom{6 + 9 - 1}{9 - 1} = \binom{14}{8} = \binom{14}{6} = 3003.$$

5 Problem solving

	Order matters	Order ignored
No repetition	$P(n, r) = n!/(n - r)!$	$\binom{n}{r}$
Repetition allowed	n^r	$\binom{r+n-1}{n-1}$

Without repetition, $0 \leq r \leq n$. With repetition, $n \geq 1$, $r \geq 0$, and each type has unlimited supply. Extra restrictions require additional steps.

5.1 A group photograph

Eight distinct students include A and B . Choose five students for a photograph and arrange them in a line. Both A and B must be included, but they may not stand next to each other. How many photographs are possible?

One outcome specifies both the five selected students and their left-to-right order. Choose the membership first, then count valid orders of that group. The remaining three members can be chosen from six students in $\binom{6}{3} = 20$ ways. For each group there are $5! = 120$ unrestricted orders. If A and B stand together, treat them as one block with two internal orders, AB and BA . The block and the three other people form four objects, giving $2 \cdot 4! = 48$ forbidden orders. For example, write the adjacent pair as one block:



Hence

$$\binom{6}{3}(5! - 2 \cdot 4!) = 20(120 - 48) = 1440.$$

Follow-up: A round table. Keep the same selection rule, but seat the five students around a round table with unlabelled seats. Rotations count as the same; reflected orders count as different. Both A and B must not be neighbours. For each selected group, there are $4!$ unrestricted circular seatings. The adjacent pair forms one block, giving four circular objects and two internal orders, hence $2 \cdot 3!$ forbidden seatings. Thus

$$\binom{6}{3}(4! - 2 \cdot 3!) = 240.$$

5.2 Committee with a restricted chair

From seven women and six men, form a five-person committee with exactly three women. The chair must be one of the selected men.

Choose the membership, then choose one of the two selected men as chair:

$$\binom{7}{3} \binom{6}{2} \cdot 2 = 1050.$$

6 Optional: A queue at the ticket counter

6.1 The problem

A ticket costs \$5. Ten customers each pay with a \$5 bill, and ten each pay with a \$10 bill. The cashier starts with no change and must serve customers in order, without skipping anyone. How many sequences of bill types allow everyone to buy a ticket? Customers with the same bill type are not distinguished.

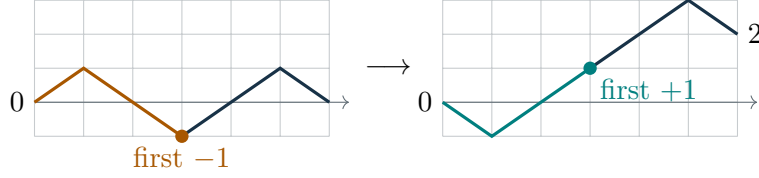
A queue succeeds exactly when every prefix (the first k customers) contains at least as many \$5-paying customers as \$10-paying customers. Each \$5 payment adds one available \$5 bill; each \$10 payment uses one for change. For example, $(5, 10, 5, 5, 10, 10)$ succeeds, but $(5, 10, 10, 5, 5, 10)$ fails at the third customer.

6.2 Paths and failed queues

Represent a \$5 payment by an up-step and a \$10 payment by a down-step. After any prefix, the height is the number of \$5 payments minus the number of \$10 payments. We need paths with ten up-steps and ten down-steps that never go below zero.

There are $\binom{20}{10}$ unrestricted paths: choose the up-step positions. We will subtract the failed paths, namely those that visit height -1 .

Transform a failed path. Find its first visit to -1 . Swap up and down in the entire initial segment ending there. Keep every later step's direction unchanged. This reflects the initial segment across height zero; the rest of the path starts two units higher.



The figure uses the smaller failed queue $(5, 10, 10, 5, 5, 10)$. The same transformation applies to twenty steps.

At the first visit to -1 , the initial segment has a up-steps and $a+1$ down-steps for some $a \geq 0$. After reflection these counts become $a+1$ and a . The whole path therefore changes from ten up-steps and ten down-steps to *eleven up-steps and nine down-steps*.

Reverse the transformation. Start with any path having eleven up-steps and nine down-steps. Its endpoint is at height 2, so it must visit $+1$. Reflect the initial segment ending at its first visit to $+1$. Before that visit, heights are at most zero; after reflection they are at least zero, and the endpoint of the segment is -1 . Thus this produces a failed path with ten steps of each kind.

For the forward transformation, the first visit to -1 becomes the first visit to $+1$. The reverse transformation therefore reflects exactly the same segment. The two operations undo each other: every failed path corresponds to exactly one path with eleven up-steps and nine down-steps.

6.3 The count

Choose the nine down-step positions of the transformed path. There is no height restriction on it, so the number of failed queues is $\binom{20}{9}$. The number of successful queues is

$$\binom{20}{10} - \binom{20}{9} = 184,756 - 167,960 = 16,796.$$

With $n \geq 1$ customers of each bill type, the same argument gives

$$\binom{2n}{n} - \binom{2n}{n-1} = \left(1 - \frac{n}{n+1}\right) \binom{2n}{n} = \frac{1}{n+1} \binom{2n}{n}.$$

This is the n th **Catalan number**. The counting idea is to replace a difficult restriction by a reversible transformation of the failed cases.

7 Practice

1. Nine runners compete for four distinct finishing places. How many results are possible, with no ties?
2. How many ways are there to choose four books from twelve distinct books?
3. Arrange seven people around a round table, identifying rotations but not reflections.
4. Count length-ten bit strings with exactly four ones.
5. Count rearrangements of ABCDEFG in which ABC stays together in any internal order.
6. Choose eight donuts from five types with unlimited supply, with repetition allowed and order ignored.

Answers

1. $P(9, 4) = 3024$.
2. $\binom{12}{4} = 495$.
3. $(7 - 1)! = 720$.
4. $\binom{10}{4} = 210$.
5. $5! \cdot 3! = 720$.
6. $\binom{5+8-1}{8} = \binom{12}{8} = 495$.